HOW WE TESTED

We divided the contestants into three major categories: entry-level dSLRs, enthusiast dSLRs and pro dSLRs. We listed all possible specifications, but only features were assigned a score and taken into account for computation of winners. For example, specifications such as sensor size and shutter speed range were listed but not scored, whereas the presence of an auto ISO mode was. For each camera, we rated the menu system and its intuitiveness, as well as the ergonomics and of the buttons and dials and their layout, and build quality.

We didn't want a camera's image quality restricted by the quality of lens. Sony didn't agree to send us the lens we wanted; but Canon, Nikon and Olympus did. For Olympus, this was the Zeiko 12-60 and Zeiko 35-100 f2.0, and for Canon and Nikon we used their 24-105 f4 IS USM and 24-120 f4 IS SWM, respectively. For Sony, we used their 18-55-kit lens.

Our image quality tests consist of focussing and subjective image quality tests. For focussing, we check each camera for front or back focussing. We focussed at a point on a diagonally placed display. A spreadsheet is displayed on the screen, since its cells serve as a good way to gauge front/back focussing. With a particular cell within focus, we observe blur and sharpness of the adjoining cells. If the sharpness is uneven further away, the camera is front focussing and if it's closer to the camera, it is a problem with back focussing. We also test each camera for wide angle and long range focussing accuracy. Here, we observe detailed objects with textures such as flowers and fruits. We do this indoors and outdoors.

In image quality, we check dynamic range and highlighting. We shoot brighter objects with dimmer and reflective backgrounds to check for spot highlighting and detail. The more overblown the detail, the lower the dynamic range of the camera's sensor. These tests are done at ISO 100 and ISO 200. For quality and micro detail, we use our test scene with a lot of colour, texture and detail. For stability, each camera is mounted on a tripod.

We use the same scene for higher ISO tests, where we keep changing the shutter speed to ensure proper exposure. For this studio test, we use a fixed aperture setting of f8. Auto white balancing is used – this also allows us to judge how well the white balance of each camera works with fluorescent light. For ISO tests, we increase ISO settings through 400, 800, 1600, 3200, 6400 and 12800. We shot in JPEG mode for most of the tests, and also used the RAW mode to check the camera's image processing.

We tested video quality at 720p and 1080p resolutions. We used zoom liberally to check for focus and clarity, and audible noise. Two 30 second clips were taken, one indoors and the other outdoors. For our flash tests, we used entry-level flashes for those dSLRs that didn't have integrated flash units.

spinning a dial and watching the on-screen menu also spin to the selected value. While this visual-based interface is novel, it isn't as intuitive as looking through a viewfinder. The sweeping panorama mode works pretty well, and is useful if you desire to capture a really wide angle view.

Overall, Sony's menu proves to be a little more visual, but also a little cumbersome to change settings quickly. Of course it could just be that we need to get used to Sony's attempt to do something new here. Nikon and Canon largely stick to their proven menu systems.

Performance

The first mini battle takes place between the Canon 1100D and Nikon D3100. The Sony A33 falls a little behind, as it struggles to resolve more detail in our test scene. This could be an issue with the lower quality lens, but to Sony's detriment they weren't willing to part with another lens after repeated attempts. The 1100D resolves more detail than the D3100. The D3100 does have a slightly more realistic texture to the orange, but overall, the 1100D appears sharper. The D5100 is clearly better than the D3100, but is very close with the Canon 1100D and 600D. Surprisingly, the 1100D holds up pretty well here, and is able to offer a brilliantly sharp image. Focussing was spot on, too.

The 600D seems to have a softer output than the 1100D, which strangely changes in the higher ISO tests, where it is able to both suppress noise and deliver more detail than the 1100D. This might be less about the sensor, and more about a stronger noise filter. However, it is outclassed by the D5100 that outperforms both Canon cameras in terms of texture.

Canon EOS 1100D 100 per cent crop at ISO 3200 - surprisingly clean output

Sony A33 100 per cent crop at ISO 3200 - note the increased noise compared to the Canon EOS 1100D